STARWEAVER®

Hands-On-Learning: AI-Powered Cybersecurity Analysis Activity

Scenario

You are a cybersecurity analyst at a mid-sized company that has been experiencing suspicious network activity. Your security team has collected various artifacts including suspicious URLs, email content, system logs, and network traffic patterns. For education purposes, as the company's AI-enhanced security specialist, you need to leverage free AI tools to analyze these artifacts and determine potential threats. This simulation will help you understand how AI agents can assist in real-time threat detection and analysis across multiple security domains.

Instructions

1 Set Up Your AI Analysis Environment

- Open ChatGPT (free version) or Claude AI in your browser
- Open VirusTotal (virustotal.com) - a free malware analysis platform
- Have a text editor ready for documenting your findings

2 Suspicious URL Analysis

Use the following suspicious URL for analysis: hxxp://secure-banking-update[.]tk/login

Defang the URL (replace 'x' with 't' and '[.]' with '.') carefully in an isolated environment

Submit the URL to VirusTotal for reputation analysis

Ask your AI assistant to analyze the URL structure and explain what makes it potentially malicious

3 System Log Anomaly Detection

Present this log snippet to your AI assistant:

```
2024-06-22 03:47:12 Failed login attempt for user 'admin' from IP 192.168.1.100
2024-06-22 03:47:15 Failed login attempt for user 'administrator' from IP 192.168.1.100
2024-06-22 03:47:18 Failed login attempt for user 'root' from IP 192.168.1.100
2024-06-22 03:47:21 Successful login for user 'admin' from IP 192.168.1.100
2024-06-22 03:47:25 File access: /etc/passwd by user 'admin'
2024-06-22 03:47:30 New user created: 'backup_admin' by user 'admin'
```

Ask the AI to identify potential security incidents and attack patterns

Request recommendations for immediate response actions

4 Malware Behavior Analysis

Use VirusTotal.com to analyze this known malware hash: d41d8cd98f00b204e9800998ecf8427e
Do not upload any real files. Use only the synthetic hash/domain above.

Ask your AI assistant to explain the behavioral patterns typical of the malware family identified

Discuss with the AI how machine learning helps in malware classification

5 Network Traffic Pattern Analysis

Present this network activity summary to your AI:

Source: 10.0.0.15 -> Multiple destinations (50+ different IPs)
Port: 80, 443, 53
Frequency: 1000+ connections per minute
Pattern: Sequential IP scanning with payload requests
Time: 2:00 AM - 4:00 AM (outside business hours)

Ask the AI to classify this network behavior and identify potential attack types

Request mitigation strategies for this type of network activity

6 Threat Intelligence Summary

- Compile your findings from all previous steps
- Use your AI assistant to create a structured threat intelligence report
- Ask the AI to prioritize the threats based on severity and likelihood
- Request recommendations for implementing AI-powered detection rules for similar future threats

7 Document Your Learning

- Create a brief summary of how AI assisted in each analysis step
- Note which types of analysis AI performed most effectively
- Identify limitations you observed in AI-assisted cybersecurity analysis

Instructions for Documenting and Sharing The Project for Peer Review

Document the Project

- Use word processing software to create your report.
- Ensure all sections of the project document structure are completed.
- Proofread and edit your report for clarity and accuracy.

Share the Project

Document: Save the report in PDF format.

Upload: Share in the “Peer Review” area of the course shell with a brief description.

Peer Review

Review the projects submitted by your peers.

Provide constructive feedback and comments on their work.